Assignments for the Dynamical Systems Course

June 23, 2026

1 Assignments set III

1.1

Show that the system

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -x + y(1 - x^2 - y^2) \end{aligned}$$

has a periodic orbit (Hint: derive the equation for the function $l(x, y) = x^2 + y^2$)

1.2

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -4x + y(1 - x^2 - y^2) \end{aligned}$$

Show that there is a periodic orbit.

1.3

$$\begin{aligned} \dot{x} &= 3x - 2y - x(x^2 + 4y^2) \\ \dot{y} &= 4x - y - y(x^2 + 4y^2) \end{aligned}$$

a) Show that (0,0) is an unstable spiral
b) Show that (0,0) is the only fixed point. (Hint: show the fixed point equations as an eigenvalue problem).
c) Let $L(x, y) = x^2/2 + y^2/2$, $M(x, y) = x^2/2 + (x - y)^2/2$. Show that $\dot{M} > 0$ for (x, y) close to $(0, 0$ and $\dot{L} < 0$ for (x, y) far from $(0, 0)$.
d) Show that there is a trapping region in the form $M(x, y) > a^2$, $L(x, y) < A^2$, where a is small and A is large.
e) Show that there is a periodic orbit in the set defined in d).

1.4

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -x - x^3 - \mu(x^4 - 1)y \end{aligned}$$

Show there is a periodic orbit for $\mu > 0$.

1.5

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -x - 2x^3 - 3x^5 + y(2\mu + x^2 + y^2) \end{aligned}$$

a) Find eigenvalues of the origin
b) For $\mu = 0$, is the origin weakly attracting or weakly repelling?
c) Show that there is a Hopf bifurcation at $\mu = 0$. Is it subcritical or supercritical?